

dose-volumetric and biological factors for predicting RP vary across reports. It has not been well studied as to how these factors from different sources are interrelated and whether such relationships can affect the predictive power of individual factors. Bayesian Network (BN) can represent variable dependencies in a directed graph and combine observations on the variables for prediction of response. Our aim was learning an interaction graph between previously known RP risk factors and evaluating its feasibility for RP prediction.

Materials/Methods: We studied 19 patients with NSCLC who received curative radiation therapy with fraction sizes 3.5 – 34 Gy (median: 16 Gy). Four RP events were observed (CTCAE grade ≥ 2). Serum concentrations of the following four candidate pro-inflammatory biomarkers were measured at baseline and end treatment: alpha-2-macroglobulin ($\alpha 2M$), angiotensin converting enzyme (ACE), transforming growth factor $\beta 1$ (TGF- $\beta 1$), and interleukin-6 (IL-6). The biomarker data was fused with dose-volumetric and clinical RP risk factors. The Koller-Sahami (KS) filter selected a subset of the variables, which joint distribution could best approximate the probability of RP. The BN was trained from the reduced dataset using the Markov Chain Monte Carlo method and enforced causal relationships. The RP predictive performance of the BN model was tested using 10-fold cross-validation.

Results: Five variables (Table 1) were chosen by the KS filter and formed a BN. The following 3 variables were found in the network to have a direct influence on RP: ACE end-to-baseline ratio, V30, and PTV location. The network found a link between V30 on ACE end/baseline in which direction was fraction size dependent: a Pearson's coefficient between the variables was 0.56 ($P = .44$) for fraction size <10 Gy and -0.33 ($P = .24$) for ≥ 10 Gy. When the BN was used as a RP classifier, the area under the ROC curve (AUC) was 0.77, which was lower than the univariate AUC for PTV volume (0.9) and V30 (0.86). Prediction using the BN was more accurate for fraction sizes ≥ 10 Gy (AUC = 0.92) than <10 Gy (AUC = 0.67).

Conclusion: The BN method identified the most important RP predictors from all three sources (biological, dose-volumetric, and clinical). Predictive power of the model depended on the dose response behavior of ACE, where contrast was shown above and below 10 Gy per fraction. This preliminary result hints at a potential pneumotoxic role of ACE in ablative dose ranges.

Author Disclosure: S. Lee: None. N. Ybarra: None. K. Jeyaseelan: None. S. Faria: None. N. Koppek: None. I. El Naqa: None.

122

Survival After Definitive Treatment for Uveal Melanoma in the Post COMS Era

Y.J. Rao,¹ S.N. Badiyan,² and P.W. Grigsby³; ¹Washington University, St. Louis, MO, ²University of Maryland, Baltimore, MD, ³Washington University School of Medicine, St. Louis, MO

Purpose/Objective(s): The Collaborative Ocular Melanoma Study (COMS) was a randomized trial, which accrued patients from 1987 to 1998, and reported similar outcomes for brachytherapy and enucleation for medium sized choroidal melanomas, and reported no added benefit of external beam radiation therapy (EBRT) to enucleation alone for large tumors. Significant advances in treatment techniques have been made since that time, and we sought to evaluate factors associated with survival for uveal melanoma in the modern era.

Materials/Methods: Case listings were downloaded using the statistical analysis software. Search criteria included all patients who received definitive treatment for uveal melanoma from 1999–2011. Patients with metastatic disease were excluded. The outcome variable analyzed was cancer specific survival. Descriptive data analyzed for patients included age at diagnosis, clinical stage, tumor size, race, surgery, local therapy, radiation, and sequence of surgery and radiation. Analyses were done in SPSS using the Kaplan-Meier technique and Cox regression for survival.

Results: Two thousand eight hundred sixty-one patients met inclusion criteria, with median follow-up of 48 months; 1,775 patients did not have a

coded tumor diameter. Two hundred sixty-five had tumors <10 mm (small), 485 were 10–15 mm (medium), and 336 were 16 mm or greater (large); 703 patients received enucleation, 1,993 received radiation, 48 received enucleation and radiation, and 117 received laser ablation and radiation. Of those who received radiation 1,508 had brachytherapy, 576 had EBRT, and 29 had brachytherapy and EBRT. Patients 60 years or older had worse 5-year cause specific survival (CSS) of 76.2% compared to those younger than 60 years 85.2% ($P < .01$). Five-year CSS of small tumors was 97.1%, medium tumors 81.2%, and large tumors 75.8% ($P < .01$). Five-year CSS for enucleation was 65.1%, brachytherapy 87.2%, and EBRT 86.8% ($P < .01$). In small and medium sized tumors, 5-year CSS of combined laser ablation and radiation therapy was 92.3%, while those of brachytherapy and EBRT were 86.8% and 86.5% respectively ($P = .03$). On Cox regression univariate analysis, age, tumor size, surgery, and radiation were significant predictors of survival. On multivariate analysis, significant contributors included older age (HR = 1.6, $P < .01$), tumor size (HR = 5.4 for medium $P < .01$, HR = 7.4 for large $P < .01$, compared to small tumors), and treatment modality (HR = 0.31 for radiation $P < .01$ and HR = 0.10 for laser and radiation $P < .01$, compared to enucleation).

Conclusion: Radiation therapy resulted in improved survival compared to enucleation even in medium and large sized tumors. Brachytherapy and external beam radiation resulted in equivalent survival. In small and medium sized tumors, combined laser ablation and radiation therapy improved survival over radiation therapy alone.

Author Disclosure: Y.J. Rao: None. S.N. Badiyan: None. P.W. Grigsby: None.

123

Clinical Outcomes With Image Guided, Notched Eye Plaque Brachytherapy for Papillary Choroidal Melanoma

J.V. Hegde, T.A. McCannel, J.M. Lamb, J. Wang, D.J. Demanes, and M. Kamrava; *University of California, Los Angeles, Los Angeles, CA*

Purpose/Objective(s): We reviewed the oncologic and visual outcomes of notched plaque brachytherapy for papillary (contiguous with the optic disc) choroidal melanoma.

Materials/Methods: We retrospectively evaluated our institution's experience from 2003–2013 and identified 41 consecutive cases of papillary choroidal melanoma treated with Iodine-125 notched plaque brachytherapy. All lesions were treated to an apical height prescription dose of 85 Gy. Notched plaque placement was confirmed using intraoperative ultrasonography. Patient demographics and tumor baseline characteristics were recorded, and outcomes evaluated included local tumor control, distant metastasis, cause-specific death, overall survival, and visual acuity.

Results: Mean age at diagnosis was 60 years. Median follow-up was 24 months (range, 1.0–117 months). Median tumor thickness by ultrasound was 3.7 mm (range, 1.0–10.2 mm), and median maximal tumor dimension was 8.7 mm (range, 4.5–17.1 mm). American Joint Committee on Cancer T-category was T1 in 57.5%, T2 in 22.5%, and T3 in 20% of patients. All lesions were contiguous with the optic disc with a mean percent tumor wrap around the disc of 44%. Crude rates of local control, distant metastasis, cause-specific death, and overall survival were 93% (38/41), 7% (3/41), 2% (1/41), and 93% (38/41), respectively. Visual acuity was categorized as $>20/40$, 20/50 to 20/200, $<20/200$ to hand motions, and

Oral Scientific Abstracts 123; Table 1 Visual Outcomes for 41 Papillary Choroidal Melanoma Patients Treated With Notched Plaque Brachytherapy

Visual acuity prior to plaque placement $>20/40$ 20/50 to 20/200 $<20/200$ to hand motions Light or no light perception	25/41 (61%) 10/41 (24%) 6/41 (15%) 0/41 (0)
Visual acuity at last follow-up $>20/40$ 20/50 to 20/200 $<20/200$ to hand motions Light or no light perception	9/41 (22%) 9/41 (22%) 15/41 (37%) 8/41 (20%)